

Lab 2: Linear Regression on a Real-World Dataset

Apply linear regression, the cost function, and gradient descent to a real-world dataset.

Select a real-world dataset from a reliable source such as Kaggle or the UCI Machine Learning Repository. Your dataset must contain:

- At least one numerical input feature.
- One numerical target variable.
- Sufficient data for training and testing.

Examples include predicting salary from experience, sales from advertising expenditure, or marks from study hours.

Requirements

Using Python, NumPy, Pandas, and Matplotlib:

1. Load and inspect the dataset.
2. Handle missing or invalid values, if any.
3. Select one feature x and one target y .
4. Plot a scatter plot showing the relationship between x and y .
5. Split the data into training and testing sets.
6. Apply feature scaling if required.
7. Implement the prediction equation: $\hat{y} = wx + b$
8. Implement the Mean Squared Error cost function manually.
9. Implement gradient calculation and gradient descent manually.
10. Train the model and plot the cost history.
11. Plot the final regression line with the data.
12. Evaluate the model on the test data using MSE and RMSE.
13. Experiment with at least two different learning rates and compare the results.

Submission

Submit one Jupyter Notebook or Google Colab notebook containing:

- Dataset source and a brief description.
- Complete code, outputs, and required plots.
- Final values of w and b .
- Test MSE and RMSE.
- A short conclusion explaining whether linear regression was suitable for the selected dataset.

Note: Do not use a ready-made linear regression model for the main implementation. The prediction, cost function, gradients, and gradient descent must be implemented manually.